respect to k (if you don't know calculus, don't worry, you can skip the next few lines and resume on the next page, where we will be using the result of this computation).

To make our job easier, we can rewrite f by applying natural logarithm and exponentiation[21], so that we get:

$$f(k) = \left(1 - e^{-\frac{k \cdot n}{m}}\right)^k = e^{k \cdot \ln\left(1 - e^{-\frac{k \cdot n}{m}}\right)}$$

This function is minimal when its exponent is minimal, so we can define function g:

$$g(k) = k \cdot \ln\left(1 - e^{-\frac{k \cdot n}{m}}\right)$$

And compute the derivatives of g instead, which is easier to work with.

The first order derivative of `g(k)` is

$$g'(k) = \frac{\partial g}{\partial k} = \ln\left(1 - e^{-\frac{k \cdot n}{m}}\right) + \frac{k \cdot n}{m} \cdot \frac{-e^{-\frac{k \cdot n}{m}}}{1 - e^{-\frac{k \cdot n}{m}}}$$

and it becomes equal to `0` for `k=ln(2)*m/n`.

To make sure this is a minimum for the function, we still need to compute the second order derivative and check that it returns a negative value when computed on the *zero* of `g'`: